Day 3: From Modelling to Operations: Supporting Day-to-Day Decisions with Delft3D FM

Welcome to the break out session "**From Modelling to Operations: Supporting Day-to-Day Decisions with Delft3D FM**"! Before we start, this guide helps you set up and launch the FEWS_DKEAU Demo System hosted at the Deltares server.

1. Download and Extract

1. Download FEWS_DKEAU.zip to your local computer: https://download.deltares.nl/delft3d-user-days-2025-day-3-break-out-session-from-modelling-to-operations
2. Unzip it to a local drive without spaces or special characters in the path, for example: C:\FEWS_DKEAU\

2. Explore the Folder Structure

Inside the FEWS_DKEAU folder, you will find:
- bin → contains the Delft-FEWS binaries
- DKEAU → contains the DKEAU connection configurations
- Start_FEWS_DKEAU_DEMO.bat à link to start the FEWS_DKEAU demonstration system.

3. Launch Delft-FEWS

1. Double-click Start_FEWS_DKEAU_DEMO.bat.
2. The batch file reads the connection configuration and connects to the Deltares server (this is a live system!).

4. Wait for Startup

You should see the startup screen. The first load may take a few minutes, depending on your internet connection.

5. Verify Successful Launch

Once complete, you should see the main view of the FEWS_DKEAU system. Yes, it looks like an empty map over Dunkirk. We will run models and see results during the exercises.

Day 3: Deep-dive in pre- and postprocessing

Installation guide

The installation guide provides you with a dedicated Python environment for the break-out session. With these environments you should be able to work through the exercises for each tool, xugrid and HYDROLIB-core.

Installation Miniforge3

Download Miniforge3 from conda-forge.org and install it with the recommended settings.

Installation Python environment

- Open "Miniforge Prompt" (via start menu)
- Run the following command in the Miniforge Prompt:
  `conda create --name dsd_env python=3.12 spyder` (spyder is optional)
- When this has finished succesfully, run:
  `conda activate dsd_env`
- Once in the active environment, install all dependencies with pip:
  `pip install xugrid hydrolib-core[examples]`
- Go to the path where you downloaded the course materials (see below)
- To run jupyter notebooks:
  `jupyter notebook`

Download required software and course materials

You will be able to download all the necessary course materials for the breakout session via the download portal: https://download.deltares.nl/delft3d-user-days-2025-day-3-break-out-session-deep-dive-in-pre-and-postprocessing